

Working Title: Say More, Mean Less: Voluntary Disclosure Under Economic Policy Uncertainty

Target Journal: Finance Research Letters

Section 1: Introduction

Purpose: Motivate the question, preview findings, and state the paper's contribution.

Structure

- Open by establishing EPU as a pervasive macro-level force with well-documented effects across corporate decisions and financial markets — investment (Gulen & Ion, 2016), M&A (Bonaime et al., 2018), bond volatility (Gong et al., 2023), lending markets (Zhou et al., 2024), and public procurement (Leite, 2025). Despite this breadth, EPU's effect on voluntary disclosure — one of the most direct channels through which managers communicate with investors — remains underexplored.
- Voluntary disclosure theory (Verrecchia, 1983; Dye, 1985) predicts that managers weigh the benefits of reducing information asymmetry against the costs of disclosure — litigation exposure, proprietary costs, and the risk of committing to statements that may prove inaccurate. EPU sharpens this trade-off: rising uncertainty raises both the demand for forward-looking information from investors and the costs of providing it. Whether managers respond by updating their disclosures or retreating to prior-period language, and whether they disclose more or less, is an empirical question.
- We examine voluntary disclosure behavior through two complementary lenses: (1) modification in forward-looking language — do managers genuinely update their disclosures year-over-year, or not? and (2) the volume of forward-looking content — do managers disclose more or less, and does any increase reflect specific, informative commitments or vague, non-committal language?
- We ask two research questions: (1) Does EPU affect the modification of voluntary forward-looking disclosures — i.e., do managers update prior-period language under heightened uncertainty? (2) Does EPU affect the volume of voluntary forward-looking disclosures?
- We find that EPU leads to less modification in voluntary disclosure — managers do not modify prior-period forward-looking language. VIX exhibits opposite effects on both dimensions. Crisis periods amplify the EPU results, and findings are robust across two independent identification methods and multiple specification checks.
- This paper contributes to three literatures. First, we extend the voluntary disclosure literature (Verrecchia, 1983; Nagar et al., 2019; Jiang et al., 2022) by showing that EPU has asymmetric effects on disclosure novelty vs. disclosure volume, a distinction obscured by studies focusing only on quantity. Second, we contribute to the EPU literature by documenting a new disclosure channel through which policy uncertainty shapes corporate communication behavior. Third, methodologically, we employ dual

uncertainty proxies (EPU and VIX) and dual forward-looking content identification methods (Li, 2010; Huang et al., 2023) to validate findings across approaches.

Key Papers: Baker et al. (2016); Verrecchia (1983); Dye (1985); Nagar et al. (2019); Jiang et al. (2022); Gulen & Ion (2016); Bonaime et al. (2018); Gong et al. (2023); Zhou et al. (2024); Leite (2025); Rao et al. (2026); Li (2010); Huang et al. (2023); Brown & Tucker (2011).

Section 2: Literature Review & Hypotheses

Purpose: Position the paper at the intersection of three literature streams and develop testable hypotheses.

Structure

Strand 1: EPU and Its Wide-Ranging Effects on Corporate and Financial Behavior

- EPU affects investment (Gulen & Ion, 2016; Julio & Yook, 2012), M&A (Bonaime et al., 2018), and financing decisions — establishing the real-options mechanism under which firms delay or adjust behavior under policy uncertainty.
- Beyond firm-level real decisions, EPU shapes financial markets broadly: bond volatility (Gong et al., 2023), Bitcoin (Wang et al., 2019), P2P lending rates and volumes (Zhou et al., 2024), and financial stress transmission (Rao et al., 2026) — motivating its extension to the disclosure channel.
- EPU also affects narrative R&D disclosures: Jung & Kim (2024) find that EPU increases both the quantity and forward-looking content of R&D disclosures in 10-Ks, providing the closest precedent for this study and establishing a direct parallel hypothesis.

Strand 2: Voluntary Disclosure Under Uncertainty

- Voluntary disclosure theory (Verrecchia, 1983; Dye, 1985) predicts a tradeoff between reducing information asymmetry and incurring proprietary or litigation costs.
- Nagar et al. (2019) and Boone et al. (2021) show EPU increases voluntary disclosure frequency; Jiang et al. (2022) show annual report length rises with EPU — consistent with managers signaling to reduce asymmetry.
- FLS are distinct from general disclosure volume: they carry litigation risk and are harder to commit to under uncertainty, making the direction of the EPU-FLS relationship non-obvious.

Strand 3: Textual Measurement of MD&A and FLS

- FL_Modify builds on Brown & Tucker (2011) and Thomas et al. (2024) for the length-adjusted VSM/TF-IDF modification measure.
- FLS identification uses Li (2010) keyword approach and Huang et al. (2023) FinBERT-FLS large language model — dual methods that allow cross-validation of results.

Hypotheses

H1: Higher EPU leads managers to recycle prior FLS language (negative FL_Modify), as committing to novel forward-looking language is riskier under elevated policy uncertainty.

H2: Higher EPU leads to more FLS in total (positive CountFLS), driven by non-specific statements, as managers increase vague forward-looking disclosure to signal engagement without specific commitments.

Key Papers: Baker et al. (2016); Verrecchia (1983); Dye (1985); Bamber & Cheon (1998); Nagar et al. (2019); Jiang et al. (2022); Boone et al. (2021); Gulen & Ion (2016); Bonaime et al. (2018); Jung & Kim (2024); Gong et al. (2023); Wang et al. (2019); Li (2010); Huang et al. (2023); Brown & Tucker (2011).

Section 3: Data & Methodology

Purpose: Describe sample construction, variable measurement, and the empirical model.

Structure

Sample: 1995-2020, U.S. public firms. MD&A text from SEC EDGAR; EPU from policyuncertainty.com (Baker et al., 2016); VIX from FRED; financial data from Compustat/CRSP. All continuous variables winsorized at 1st/99th percentiles.

Dependent Variables

- FL_Modify (Li) and FL_Modify (Huang et al.): VSM/TF-IDF modification measures adjusted for MD&A length via 5th-order polynomial following Brown & Tucker (2011) and Thomas et al. (2024).
- CountFLS variants: total FLS count (Li, 2010; Huang et al., 2023), specific FLS count, and non-specific FLS count.

Key Independent Variables

- LN(EPU): Natural log of 12-month average of Baker et al. (2016) news-based Economic Policy Uncertainty index, calculated over each firm's financial year.
- LN(VIX): Natural log of 12-month average CBOE Volatility Index as an alternative market-based uncertainty proxy, allowing comparison of policy uncertainty vs. market uncertainty effects (cf. Wang et al., 2019; Gong et al., 2023).

Controls

- Firm-level: |DeltaStockLiquidity|, |DeltaEPS|, |DeltaCurrentRatio|, |DeltaDebtDue|, |DeltaLeverage|, |DeltaFCF|, |DeltaStd(Return)|, Downsize, Acquire, Size, Big6, Num8kFilings, HHI, M/B, OANCF.

- Macro-level: Recession, Presidential Election, Real GDP Growth, StockSentiment, JLN_MacroUncertainty, GDPDispersion.

Model: OLS with firm fixed effects; standard errors clustered at the firm level. Table 1 (descriptive statistics) and Table 2 (correlation matrix, forthcoming) are presented in this section.

Key Papers: Baker et al. (2016); Li (2010); Huang et al. (2023); Brown & Tucker (2011); Thomas et al. (2024); Jurado, Ludvigson & Ng (2015); Hope & Liu (2022).

Section 4: Main Results

Purpose: Present and interpret the primary empirical findings across three main tables.

Structure

4.1 EPU and FLS Modification (Table 3)

- EPU is negatively associated with FL_Modify across both Li and Huang et al. measures — higher policy uncertainty leads firms to less modification of prior forward-looking language rather than update it, consistent with H1.
- VIX has the opposite positive sign — market volatility is associated with more modification, suggesting policy uncertainty and market uncertainty operate through different channels (echoing the divergent EPU vs. VIX dynamics in Gong et al., 2023 and Wang et al., 2019).
- When both EPU and VIX are included jointly, the divergence strengthens — the two proxies are not substitutes and capture distinct dimensions of uncertainty.
- Mechanism: under policy uncertainty, managers face higher litigation risk from additional FLS and greater uncertainty about future commitments, leading to conservative recycling of prior language; VIX-driven volatility may instead pressure managers to communicate change actively given volatility in the equity market.

4.2 EPU and FLS Count (Table 4)

- EPU is positively associated with the FLS count and non-specific FLS, but has no significant effect on specific FLS — managers add more vague forward-looking language under policy uncertainty, not more precise commitments, consistent with H2.
- Parallels Jung & Kim (2024): just as EPU increases quantity and forward-looking content of R&D disclosures, EPU increases FLS quantity here — but the specificity breakdown adds nuance: the increase is driven by non-committal language.

4.3 VIX and FLS Count (Table 5)

- VIX is negatively associated with FLS count across all measures, market volatility reduces the quantity of forward-looking statements, the opposite of EPU's effect on count.
- This asymmetry between EPU and VIX on both dimensions (modification and count) is a key contribution of the paper.

Key Papers: Baker et al. (2016); Li (2010); Huang et al. (2023); Verrecchia (1983); Nagar et al. (2019); Jung & Kim (2024); Gong et al. (2023); Wang et al. (2019).

Section 5: Additional Analyses

Purpose: Validate and extend the main results through crisis analysis, robustness tests, and potential cross-sectional extensions.

Structure

5.1 Crisis Period Analysis (Table 6)

- Crisis periods (DotCom bust 2001-2003; Global Financial Crisis 2008-2010) are associated with significantly higher FL_Modify — during peak policy uncertainty episodes, managers do update FLS language more, providing event-based validation of H1.
- Consistent with Gong et al. (2023), who find EPU spillover effects on bond volatility are strongest during crisis periods; and with Rao et al. (2026), who show financial stress shocks have concentrated, immediate effects on EPU itself.

5.2 Robustness — Raw FL_Modify (Table 7)

- Results hold using the unadjusted FL_Modify measure with Ln(MDA_Length) added as a direct control — rules out that the polynomial length adjustment is mechanically driving results (Brown & Tucker, 2011; Thomas et al., 2024).

5.3 Potential Additional Analyses (Ekene Please suggest any other tests you think will strengthen the paper.)

- Interaction between EPU and firm-level information demand (return volatility, business cycle stage) following Jung & Kim (2024) cross-sectional design.

Key Papers: Gong et al. (2023); Rao et al. (2026); Brown & Tucker (2011); Thomas et al. (2024); Jung & Kim (2024).

Section 6: Conclusion

Purpose: Summarize findings, state implications for investors and regulators, acknowledge limitations, and suggest future research directions.

Structure

- Recap: EPU reduces FL modification (firms recycle language) but increases FLS quantity via non-specific statements; VIX has opposing effects on both dimensions; crisis periods validate the EPU mechanism.
- Situate findings within the broader EPU literature: just as EPU shapes bond markets (Gong et al., 2023), cryptocurrency (Wang et al., 2019), lending (Zhou et al., 2024), and R&D narratives (Jung & Kim, 2024), it also shapes how managers communicate the future in their MD&A disclosures.
- Implications for investors and analysts: during high-EPU periods, MD&A FLS contain more text but carry less informational novelty — readers should discount the quantity increase and focus on whether language is materially updated.
- Implications for regulators: EPU-driven disclosure recycling may create systematic opacity in mandatory filings precisely when uncertainty is highest and information demand is greatest.
- Limitations: U.S. annual reports only; cannot fully disentangle supply vs. demand for disclosure; no causal identification strategy beyond firm fixed effects.
- Future research: international settings, earnings call transcripts, EPU's interaction with firm-level information demand, and downstream capital market consequences of FLS modification.

Appendix

- Appendix A: Variable Definitions (already drafted).
- Additional robustness tables as needed.

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